

UNIVERSITY OF SOUTHAMPTON
Faculty of Engineering and Physical Sciences
School of Electronics and Computer Science

A project report submitted for the award of
MEng Computer Science With Cybersecurity

Supervisor: Dr BooJoong Kang
Examiner: ???

**A Reproducible Evaluation of
Linux Rootkit Detectors via the
Implementation of an Automated
Testbed & Proof-of-Concept
Rootkit**

by **Elliott Jelbert**
October 7, 2026

UNIVERSITY OF SOUTHAMPTON

ABSTRACT

FACULTY OF ENGINEERING AND PHYSICAL SCIENCES
SCHOOL OF ELECTRONICS AND COMPUTER SCIENCE

A project report submitted for the award of MEng Computer Science With Cybersecurity

by Elliott Jelbert

This work is all about . . .

Contents

List of Symbols	xiii
Acknowledgements	xv
1 Background & Literature Review	1
1.1 Rootkits: Definition & Context	1
1.2 Existing Rootkit Techniques	1
1.3 Rootkit Detection Techniques	2
1.4 Analysis of Previous Evaluation Methodologies	2
Bibliography	3

List of Figures

List of Tables

Listings

List of Symbols

w The weight vector

Acknowledgements

Thanks to no one.

To ...

Chapter 1

Background & Literature Review

1.1 Rootkits: Definition & Context

Rootkits are typically described as a specialised piece of malware built for persistence & stealth which is installed *after* an attacker has gained root-level access to a system ([National Institute of Standards and Technology, n.d.](#)). They may also give an attacker additional functionality such as network backdoors (cite magic bpf), the ability to hide files or other capabilities that give the attacker a greater ability to hide their presence whilst maintaining persistence.

Linux & subsequently Linux rootkits have been chosen for this project due to its relevance and prevalence in cloud & server environments. Additionally, a wide range of rootkit techniques exist for Linux which will be covered in the following section. Consequently, many tools have been developed which aim to detect the presence of a rootkit on a system. To achieve this, these tools may take a variety of approaches which will be covered in section 1.3.

The reliability and accuracy of these tools against current rootkits is paramount to ensure that...

1.2 Existing Rootkit Techniques

Linux rootkits have had an evolution

1.3 Rootkit Detection Techniques

1.4 Analysis of Previous Evaluation Methodologies

This is my literature review. Testing testing [Smith \(2024\)](#).

Bibliography

National Institute of Standards and Technology. **Rootkit**. Computer Security Resource Center Glossary, n.d. Accessed: 07-10-2026.

Jane Smith. A paper title. *Journal Name*, 12:1–10, 2024.